SURICATA INSTALLATION,CONFIGURATION AND DETECTION ON KALI LINUX

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Date:-30/07/2026

1. Introduction

Suricata is an open-source network security tool used to monitor network traffic and detect suspicious activity using security rules. In this practical, Suricata was installed and tested on a Kali Linux virtual machine.

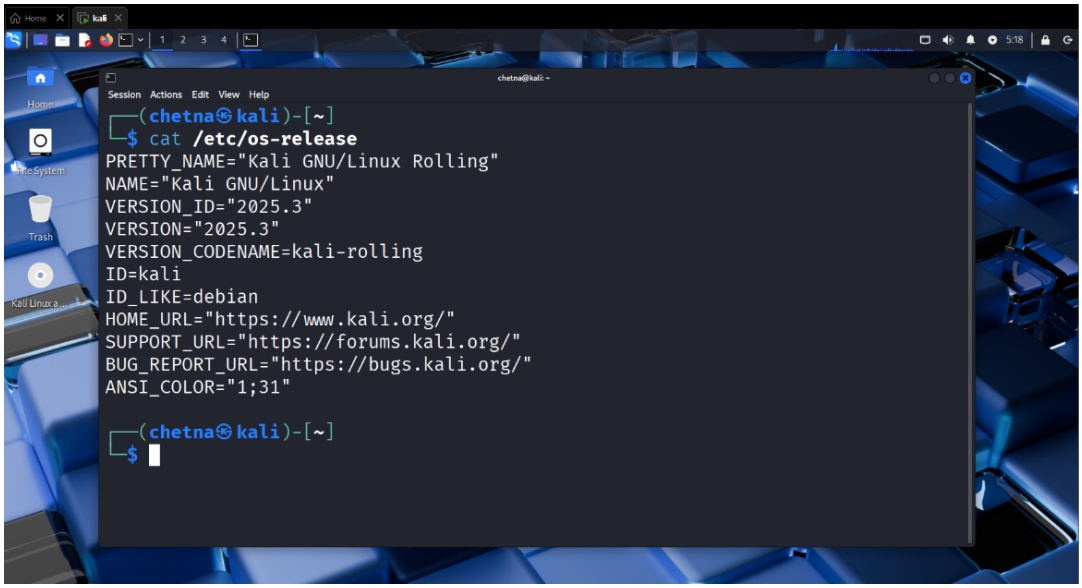
2. Objective

The objective of this practical is to install Suricata, configure its detection rules, monitor network traffic, generate controlled test traffic, and verify security alerts.

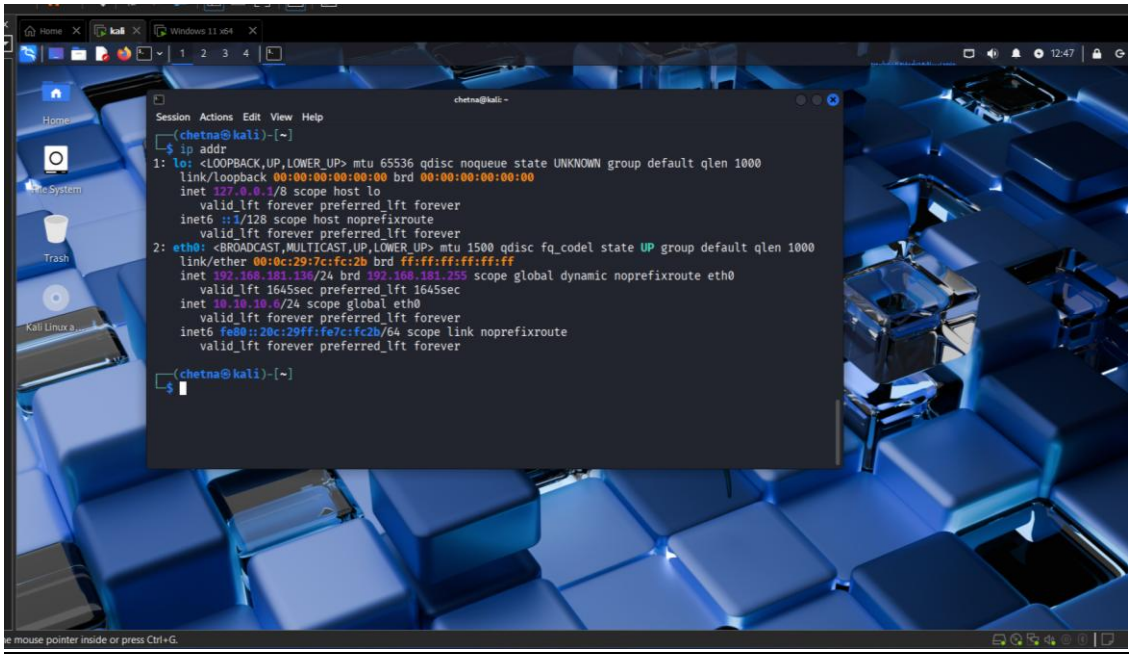
3. Lab Environment

Item	Details
Operating System	Kali Linux
Target System	Windows 11 VM
Environment	Virtual Machine
Security Tool	Suricata
Mode	IDS
Log Location	/var/log/suricata/
Detection Tool	Nmap

4. kali linux os Information

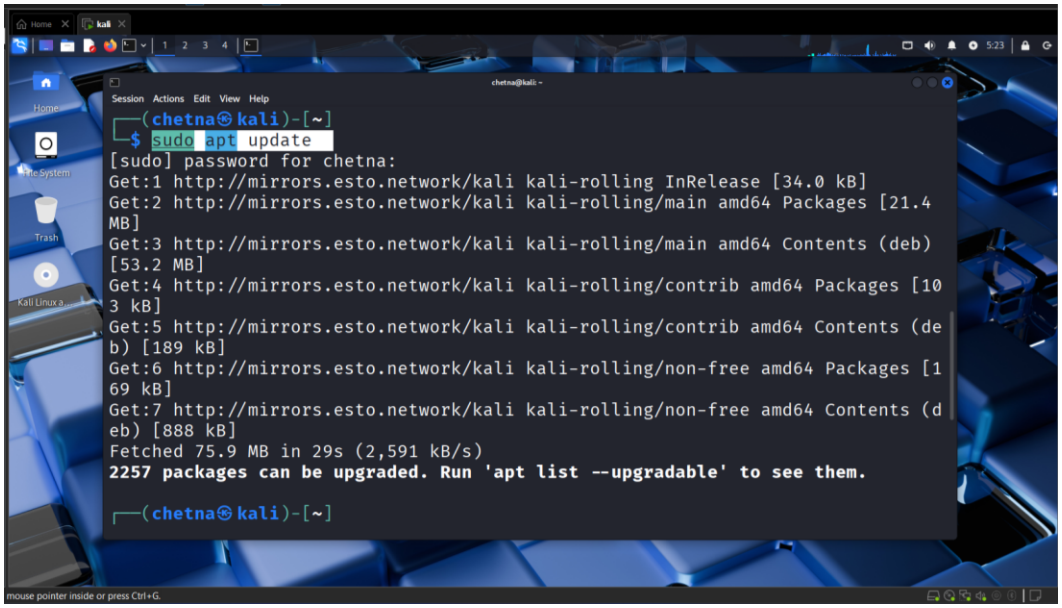


5.Network Interface And ip address configuration:-

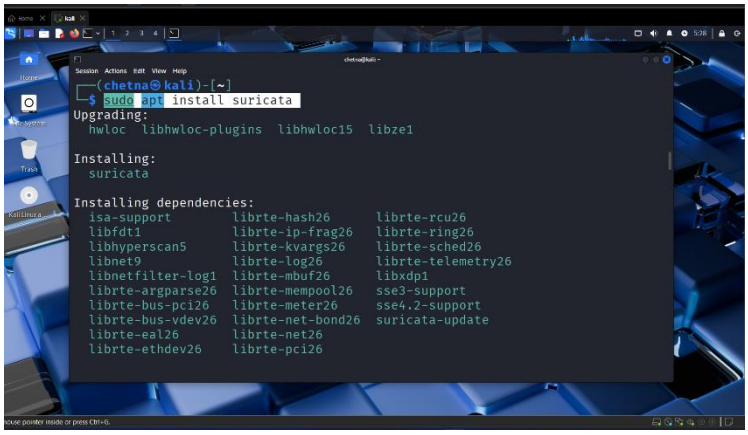


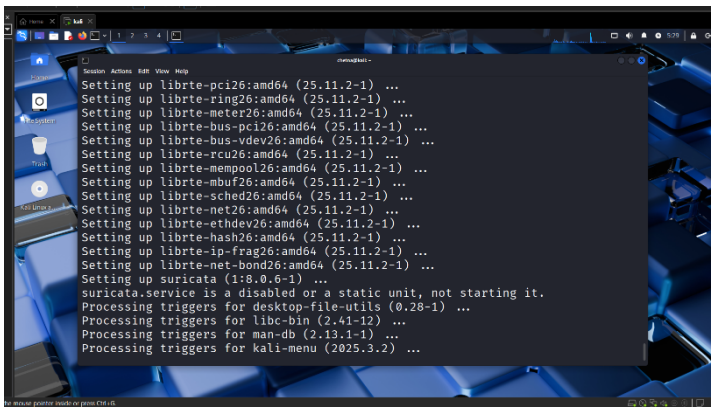
6. Update Package Repository

Before installing Suricata, the Kali Linux package repository was updated to ensure that the latest available package information was used.



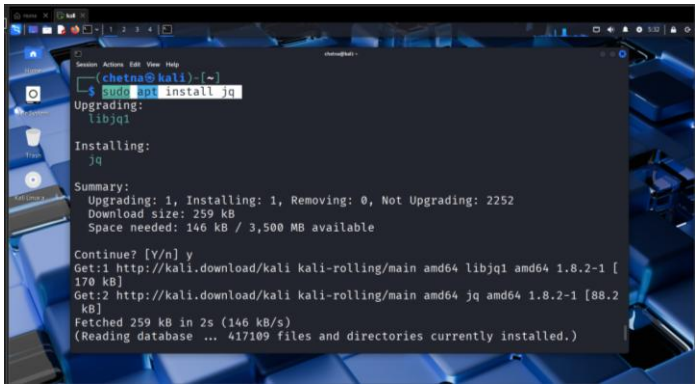
7. Install Suricata[Suricata was installed using the kali linux APT package manager.]

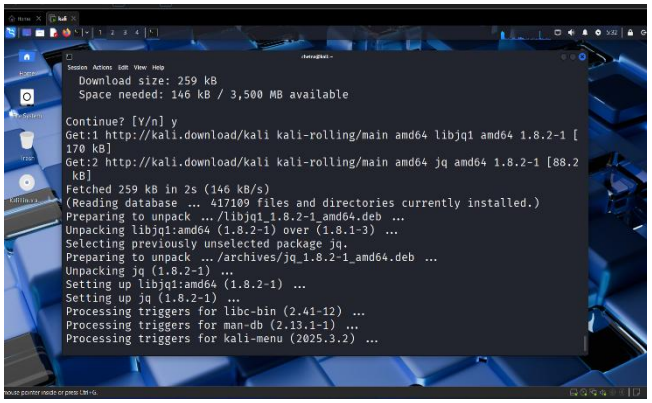




8. Install jq

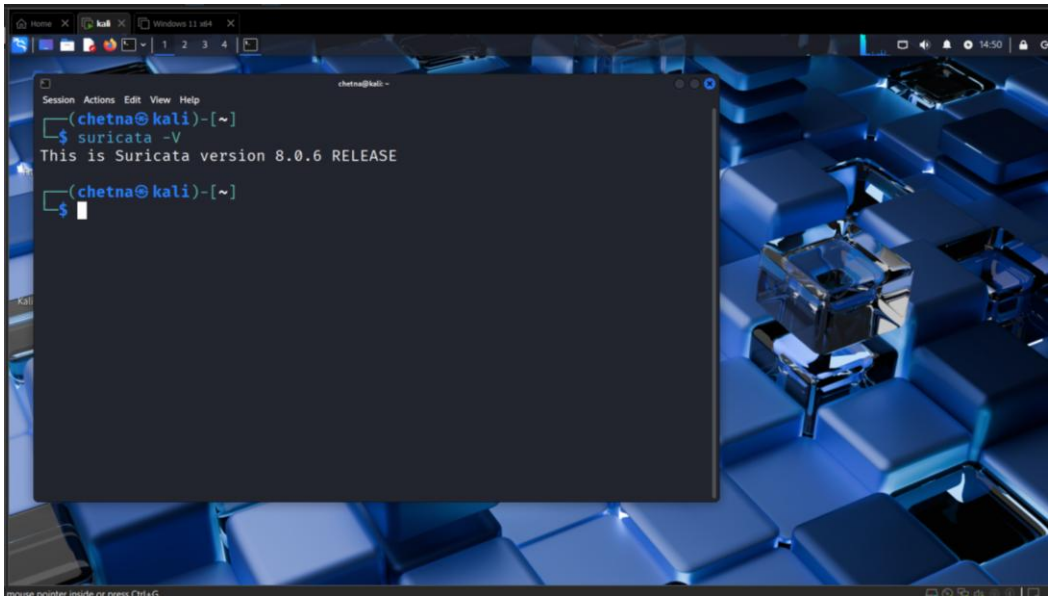
The jq utility was installed to make JSON-based Suricata logs easier to read and analyze





9.verify suricata installation

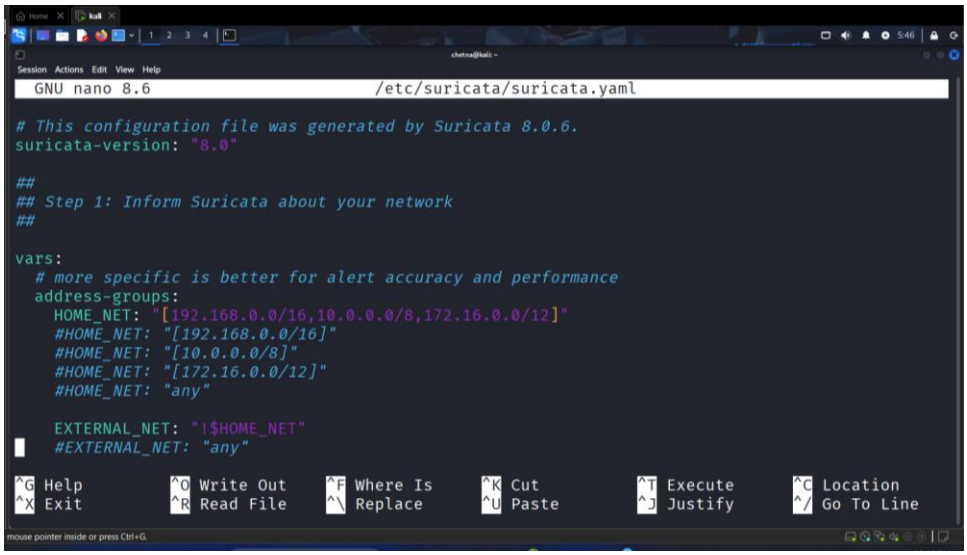
After installation, the suricata version was checked to verify that the installation was successful.



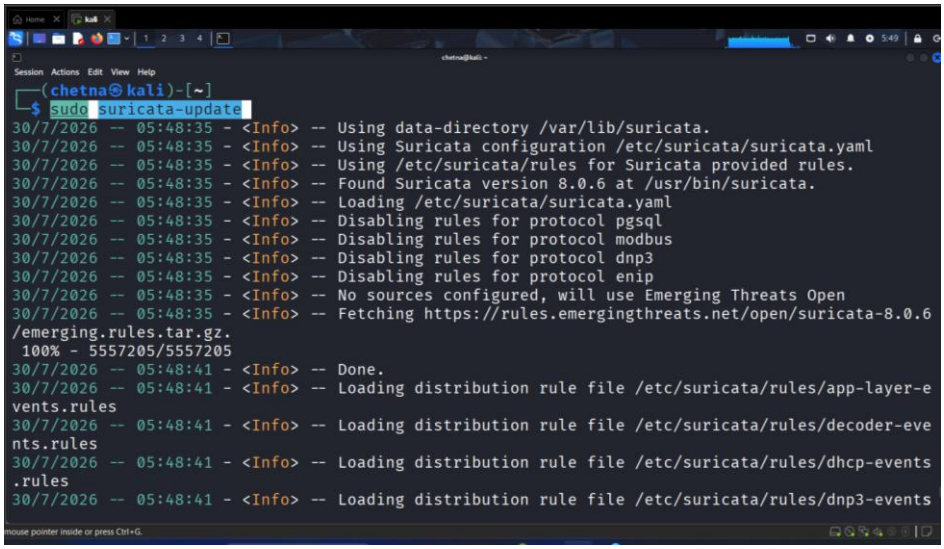
10. Suricata configuration file

Main file is: /etc/suricata/suricata.yaml

Command: sudo nano /etc/suricata/suricata.yaml

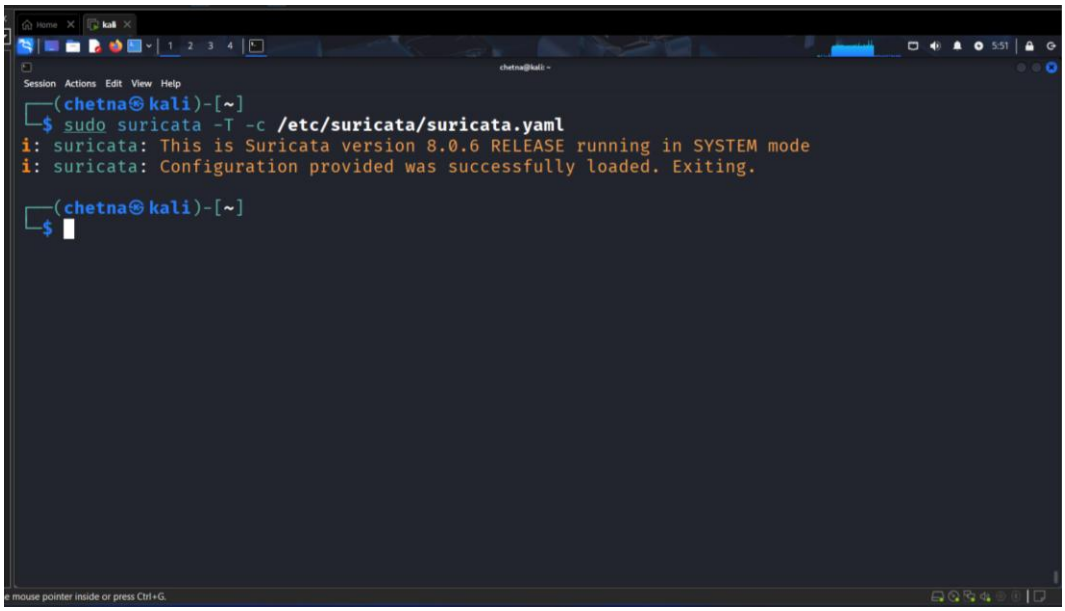


11. Update Suricata Rules

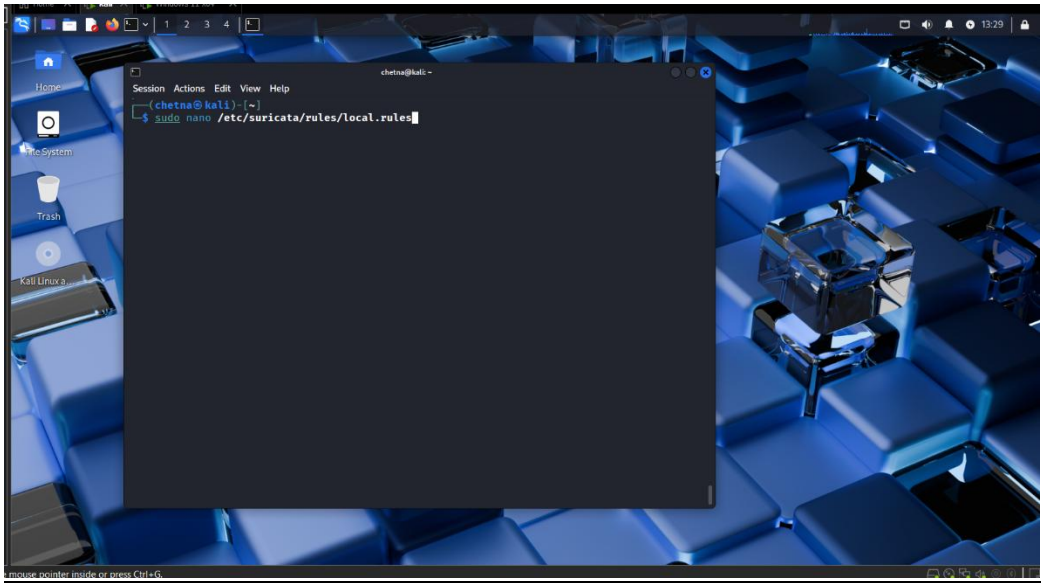


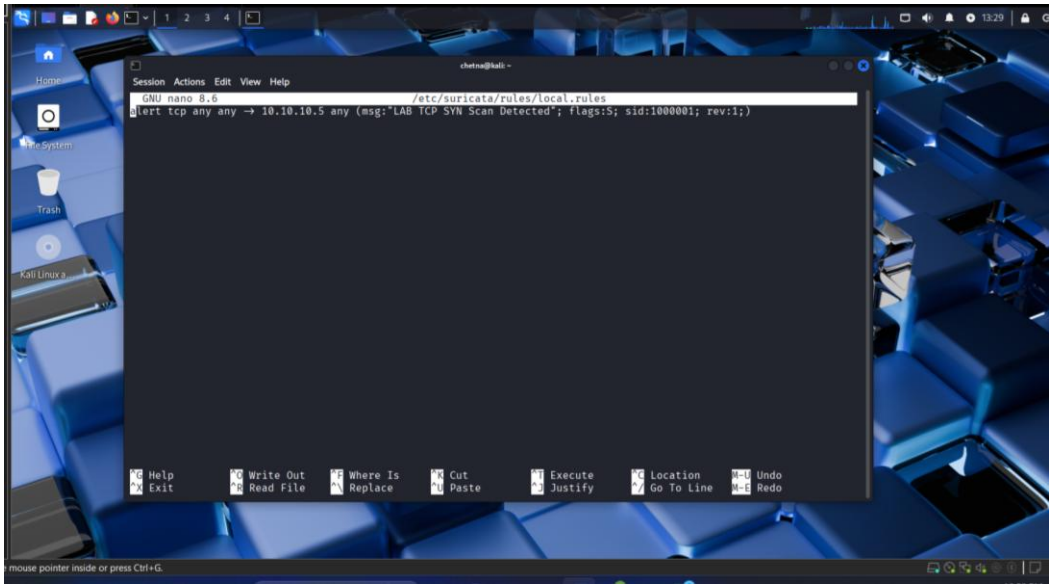
12.Test suricata configuration

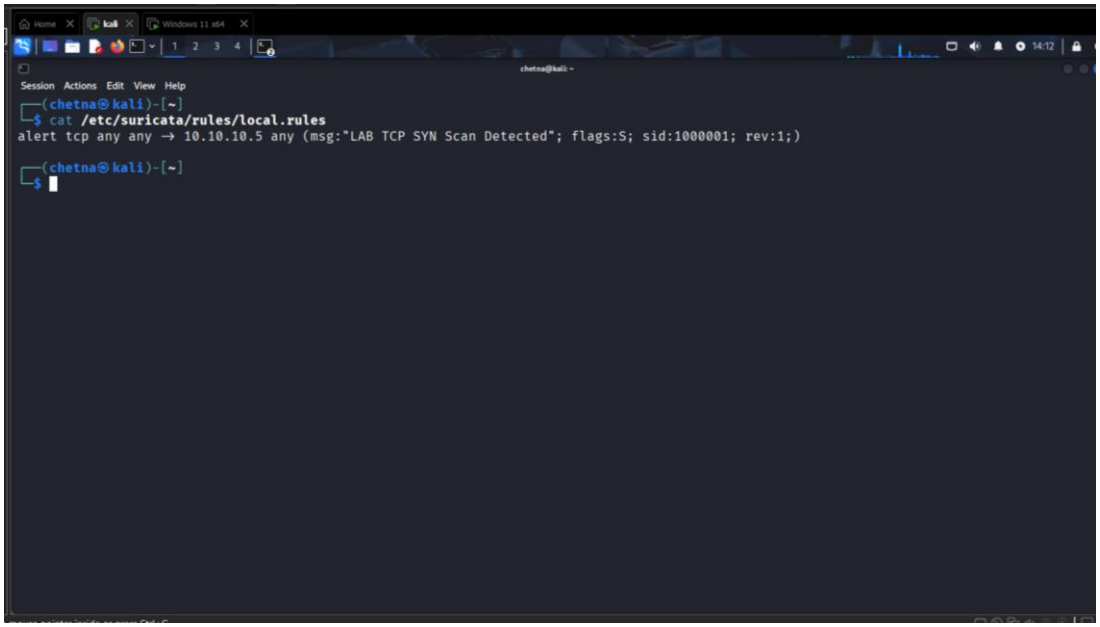
Before starting Suricata, the configuration and rules were tested to ensure that there were no configuration errors.



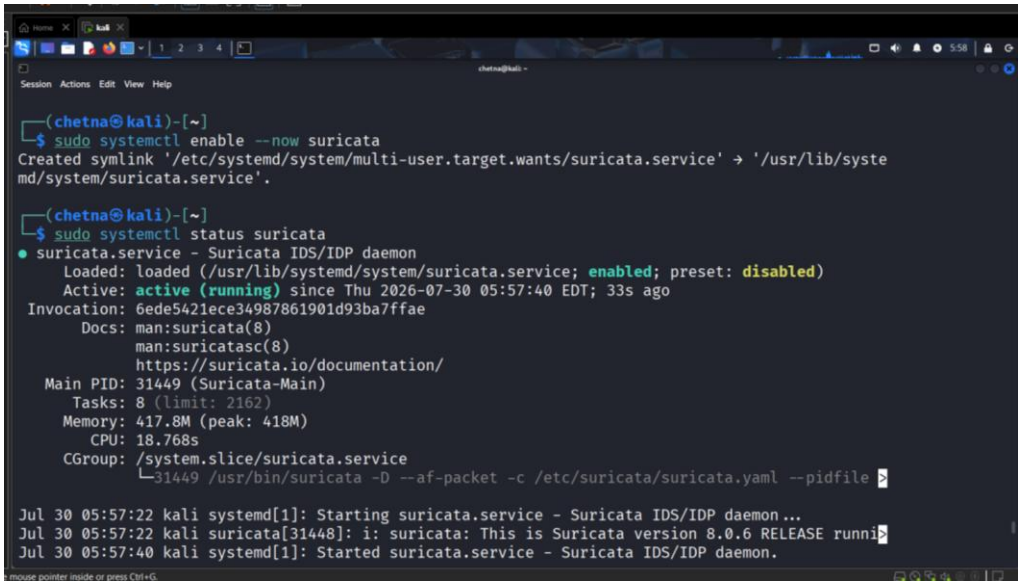
13. Custom Detection Rule


Session Actions Edit View Help
(chetna㉿kali)-[~]
$ sudo nano /etc/suricata/rules/local.rules

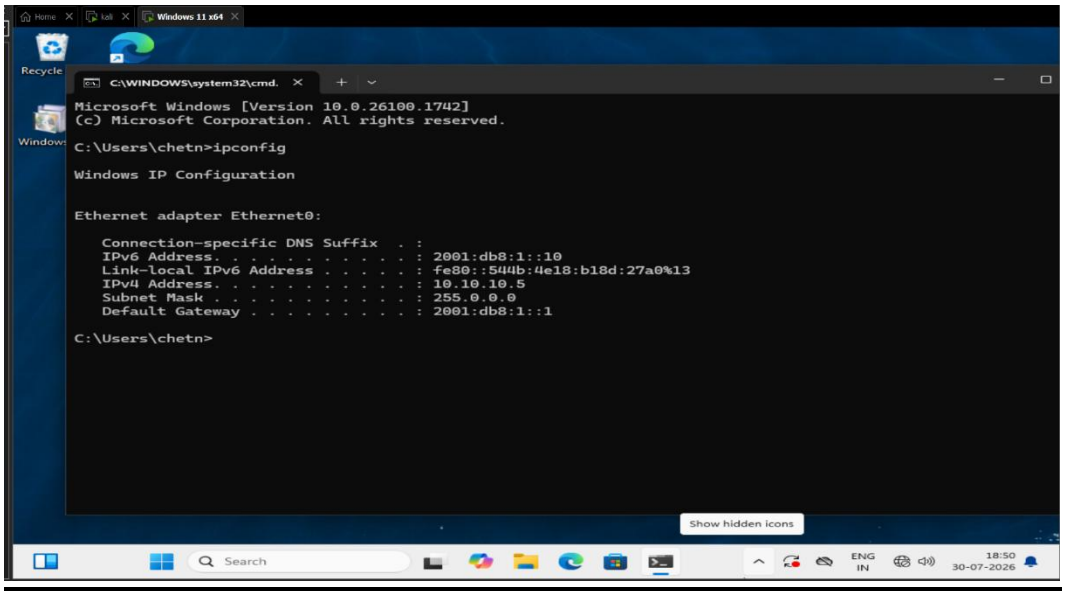

Session Actions Edit View Help
GNU nano 8.6 /etc/suricata/rules/local.rules
alert tcp any any → 10.10.10.5 any (msg:"LAB TCP SYN Scan Detected"; flags:S; sid:1000001; rev:1;)
Help
Write Out
Where Is
Cut
Execute
Location
Undo
Exit
Read File
Replace
Paste
Justify
Go To Line
Redo

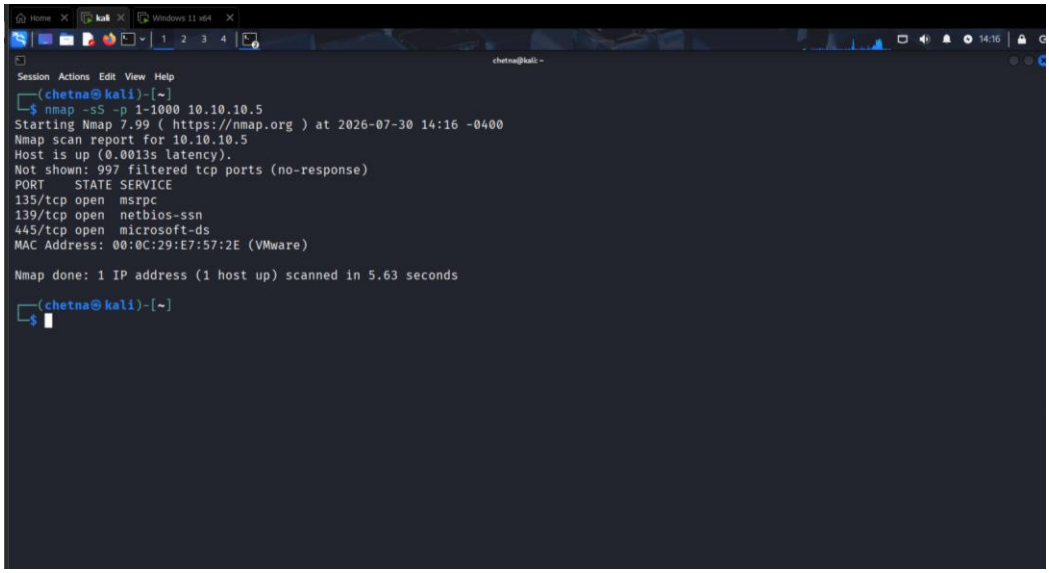

Session Actions Edit View Help
(chetna㉿kali)-[~]
$ cat /etc/suricata/rules/local.rules
alert tcp any any → 10.10.10.5 any (msg:"LAB TCP SYN Scan Detected"; flags:S; sid:1000001; rev:1;)
(chetna㉿kali)-[~]
$

14.Start Suricata

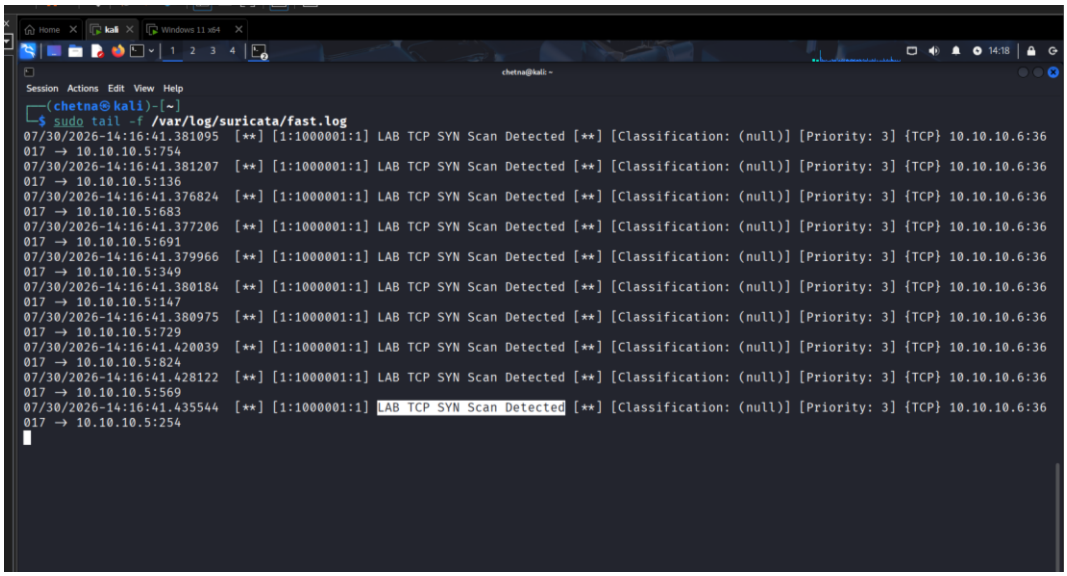


15. Windows 11 Target verification

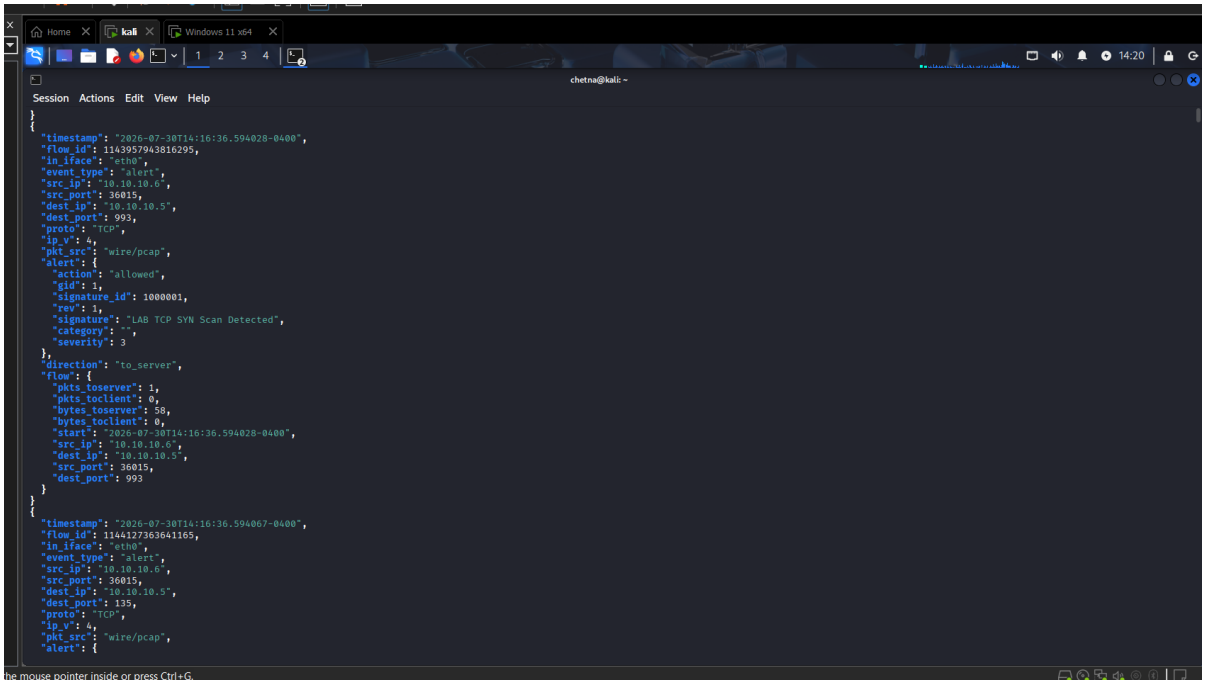




17. Monitor Suricata Alerts



Command:- **sudo tail -f /var/log/suricata/eve.json | jq 'select(.event_type=="alert")'**



19. Alert Analysis

The detected alert can be analyzed using the source and destination information.

- Source IP: Kali Linux VM
- Destination IP: Windows 11 VM (10.10.10.5)
- Protocol: TCP
- Detection: TCP SYN traffic
- Signature: LAB TCP SYN Scan Detected

The alert confirms that Suricata inspected the network traffic and matched the traffic against the custom detection rule.

20. Result

The Suricata IDS was successfully installed and configured on Kali Linux. A Windows 11 virtual machine was used as the controlled target. A TCP SYN scan was generated using Nmap and Suricata was configured to detect the traffic using a custom rule. The resulting alert was verified through fast.log and EVE JSON logs.